

Curriculum Vitae

Sarah Morell

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ACADEMIC POSITIONS

- **Researcher** (with tenure track to Senior Researcher)
University of Bremen, Germany Oct 2025 – present
- **Postdoctoral Researcher**
University of Bremen, Germany Apr 2025 – Sep 2025
Supervisor: Prof. Dr. Nicole Megow
- **Research Assistant / PhD Student**
Technische Universität Berlin, Germany Sep 2018 – Mar 2025
Supervisor: Prof. Dr. Martin Skutella
PhD Thesis: *Flow-based Problem Solving in Combinatorial Optimization*
Grade: Magna cum laude

EDUCATION

- **PhD in Mathematics**
Technische Universität Berlin, Germany Sep 2018 – Mar 2025
Supervisor: Prof. Dr. Martin Skutella
Thesis: *Flow-based Problem Solving in Combinatorial Optimization*
Grade: Magna cum laude
- **Master in Mathematics, Minor in Computer Science**
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland Sep 2015 – Jul 2018
Master Thesis: *Algorithms for Diversity Maximization*
Supervisor: Prof. Dr. Friedrich Eisenbrand
- **Bachelor in Mathematics**
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland Sep 2009 – Feb 2014
- **Abitur (High School Diploma)**
Gymnasium Eckhorst, Bargteheide, Germany Jul 2008

PUBLICATIONS

- 2026 **S. M.**, S. Debgupta, M. Skutella
Unsplittable Transshipments
 In *Proceedings of ICALP 2026*.
- 2025 **S. M.**
Flow-based Problem Solving in Combinatorial Optimization
 PhD Thesis, Technische Universität Berlin.
- 2025 **S. M.**, E. Bamas, L. Rohwedder
The Submodular Santa Claus Problem
 In: *Proceedings of SODA 2025*.
- 2022 **S. M.**, M. Skutella
Single Source Unsplittable Flows with Arc-wise Lower and Upper Bounds
 In: *Mathematical Programming*, 195(1), 1–35.
 Extended abstract: *IPCO 2020*.
- 2021 **S. M.**, I. Seidel, S. Weltge
Minimum-cost Integer Circulations in Given Homology Classes
 In: *Proceedings of SODA 2021*.
- 2018 **S. M.**, A. Cevallos, F. Eisenbrand
Diversity Maximization in Doubling Metrics
 In: *Proceedings of ISAAC 2018*.

EVENTS AND TALKS

- 05/2026 Hausdorff Research Institute for Mathematics, Bonn, Germany
 Follow-Up Workshop of the Trimester Program "Discrete Optimization"
 Talk: *Unsplittable Transshipments*
- 01/2026 CNRS Centre Paul Langevin, Aussois, France
 Workshop "Combinatorial Optimization" (by invitation)
- 09/2025 Institut d'Études Scientifiques de Cargèse (IESC), France
 Workshop "Combinatorial Optimization" (by invitation)
 Talk: *Unsplittable Transshipments*
- 11/2024 Max Planck Institute for Informatics, Saarbrücken, Germany
 Talk: *The Submodular Santa Claus Problem*
- 09/2024 École Polytechnique Fédérale de Lausanne, Switzerland
 Summer School "Synergies of Combinatorics and Theoretical Computer Science"
- 01/2024 CNRS Centre Paul Langevin, Aussois, France
 Workshop "Combinatorial Optimization" (by invitation)
- 07/2022 École Polytechnique Fédérale de Lausanne, Switzerland
 Summer School and Workshop "Modern Trends in Combinatorial Optimization" (workshop by invitation)
- 07/2022 FUTURIUM, Berlin, Germany
 German-wide event "The 7 Greatest Mathematical Adventures"
 Public introductory talk: *P versus NP* (with Prof. Dr. Martin Skutella)

12/2021	Hausdorff Research Institute for Mathematics, Bonn, Germany Workshop "Parametrized Complexity and Discrete Optimization" as part of the Trimester Program "Discrete Optimization"
11/2021	Hausdorff Research Institute for Mathematics, Bonn, Germany Workshop "Approximation and Relaxation" as part of the Trimester Program "Discrete Optimization" Talk: <i>Unsplittable Flows and Their Application in Machine Scheduling</i>
11/2021	Mathematisches Forschungszentrum Oberwolfach, Germany Workshop "Combinatorial Optimization" (by invitation) Talk: <i>Single Source Unsplittable Flows</i>
09/2020	Technical University Chemnitz, Germany (online) DMV Annual Meeting Talk: <i>Single Source Unsplittable Flows</i>
02/2020	Schloss Dagstuhl, Leibniz Center for Informatics, Germany Seminar "Scheduling" (by invitation)
01/2020	CNRS Centre Paul Langevin, Aussois, France Workshop "Combinatorial Optimization" (by invitation) Talk: <i>Single Source Unsplittable Flows</i>
11/2019	Technical University Munich, Germany Talk: <i>Single Source Unsplittable Flows</i>
05/2019	University of Michigan, Ann Arbor, USA IPCO Summer School
03/2019	École Polytechnique Fédérale de Lausanne, Switzerland Talk: <i>Single Source Unsplittable Flows</i>
12/2018	Simons Institute for the Theory of Computing, Berkeley, USA Reunion Workshop "Bridging Continuous and Discrete Optimization"
11/2018	Mathematisches Forschungszentrum Oberwolfach, Germany Workshop "Combinatorial Optimization" (by invitation)
08 – 12/2017	Simons Institute for the Theory of Computing, Berkeley, USA Program "Bridging Continuous and Discrete Optimization"

RESEARCH STAYS

02–03/2023	Maastricht University, Netherlands Visiting Prof. Dr. Lars Rohwedder
03–04/2020	Technical University Munich, Germany Visiting Prof. Dr. Stefan Weltge

TEACHING EXPERIENCE

University of Bremen, Germany

- **Operations Research** (Bachelor, 80 participants)
Course Instructor, Winter 2025
- **Seminar on Metaheuristics** (Bachelor, 9 participants)
Course Instructor, Summer 2026

- **Algorithmic and Discrete Mathematics (ADM)** (Bachelor, 5 participants)
Teaching Assistant, Summer 2025

Technische Universität Berlin, Germany

- **Introduction to Linear and Discrete Optimization (ADM I)** (Bachelor, 50 participants)
Teaching Assistant, Winter 2024
- **Discrete Optimization (ADM II)** (Master, 20 participants)
Teaching Assistant, Summer 2022
- **Computer-Oriented Mathematics** (Bachelor, 250 participants)
Teaching Assistant (including plenary tutorial), Winter 2019
Teaching Assistant, Winter 2020, Summer 2020, Summer 2019
- **Stochastics for Computer Science** (Bachelor, 600 participants)
Teaching Assistant, Summer 2024
- **Analysis I and Linear Algebra** (Bachelor, 1000 participants)
Teaching Assistant, Winter 2022
- **Linear Algebra** (Bachelor, 800 participants)
Teaching Assistant (including plenary tutorial), Winter 2018

École Polytechnique Fédérale de Lausanne, Switzerland

- **Advanced Linear Algebra** (Bachelor, 80 participants)
Teaching Assistant, Summer 2018
- **Discrete Optimization** (Bachelor, 40 participants)
Teaching Assistant, Summer 2018

PROGRAMMING AND LANGUAGE SKILLS

- **Programming:** Python (working knowledge), C/C++ (basic knowledge)
- **Languages:** German (native), English (fluent), French (fluent)